

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9648/02

Paper 2 Core Paper

15 September 2014

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

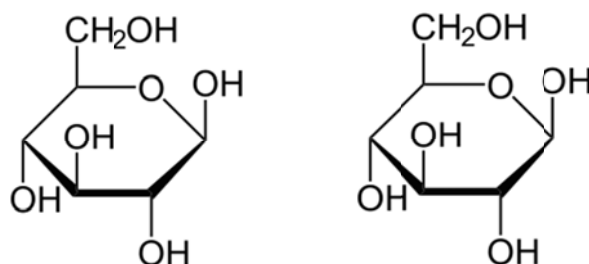
For Examiner's Use	
Section A	
1	13
2	15
3	15
4	10
5	12
6	15
Section B	
	20
Total	100

This document consists of **16** printed pages.



Section AAnswer **all** questions.

- 1 Fig. 1.1 shows two monosaccharide units that are linked by cellulose synthase to form cellulose in plant cells.

**Fig. 1.1**

- (a) (i) With reference to Fig. 1.1, name and illustrate the formation of bond between the two monosaccharide units.

[2]

name of bond [1]

- (ii) Describe **two** structural features that enable cellulose to perform its structural function in plants.

.....

 [2]

- (i) State the type of reaction catalysed by cellulase, and explain how cellulase lowers the activation energy of the breakdown of cellulose.

Describe and explain how the monosaccharide units are taken into the gut cells by facilitated diffusion.

(iii) Suggest how the gut cell can reduce the number of GLUTs in its cell surface membrane.

_____ [1]

[Turn over

- 2 Fig. 2.1 shows ribosomes attached to the endoplasmic reticulum, carrying out insulin synthesis.

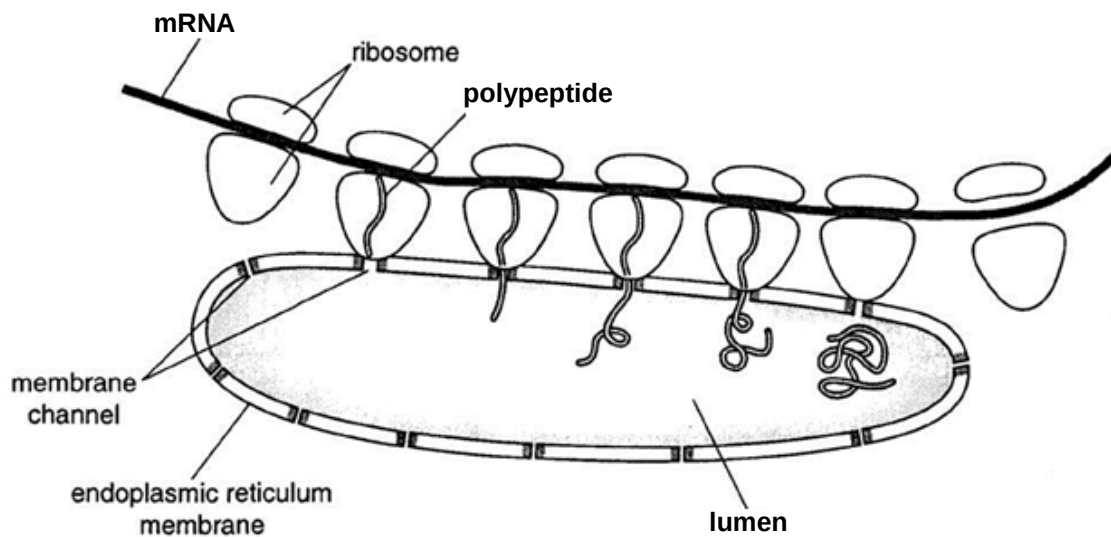


Fig. 2.1

- (a) The polypeptide entering the lumen is processed before it leaves the endoplasmic reticulum. Describe the post-translational processing of the polypeptide to form insulin.

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.....

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..... [2]

- (b) Insulin synthesised is secreted into the bloodstream. Presence of insulin in the bloodstream lowers blood glucose level. Thus production of insulin is highly regulated.

- (i) Briefly describe the events leading up to the secretion of insulin after its processing in the lumen.

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..... [3]

- (ii) Suggest how expression of the insulin gene can be regulated at translation level.

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..... [2]

Fig. 2.2 shows the sequence of bases in a section of the mRNA in Fig. 2.1, coding for the **first few amino acids** of a polypeptide chain.

3' AUGUUGCCCCCUUAGGUA AU 5'

Fig. 2.2

Each codon codes for a specific amino acid as illustrated in Table 2,

Table 2

UUU	phe	UCU	ser	UAU	tyr	UGU	cys
UUC	phe	UCC	ser	UAC	tyr	UGC	cys
UUA	leu	UCA	ser	UAA	STOP	UGA	STOP
UUG	leu	UCG	ser	UAG	STOP	UGG	trp
CUU	leu	CCU	pro	CAU	his	CGU	arg
CUC	leu	CCC	pro	CAC	his	CGC	arg
CUA	leu	CCA	pro	CAA	gln	CGA	arg
CUG	leu	CCG	pro	CAG	gln	CGG	arg
AUU	ile	ACU	thr	AAU	asn	AGU	ser
AUC	ile	ACC	thr	AAC	asn	AGC	ser
AUA	ile	ACA	thr	AAA	lys	AGA	arg
AUG	met	ACG	thr	AAG	lys	AGG	arg
GUU	val	GCU	ala	GAU	asp	GGU	gly
GUC	val	GCC	ala	GAC	asp	GGC	gly
GUA	val	GCA	ala	GAA	glu	GGA	gly
GUG	val	GCG	ala	GAG	glu	GGG	gly

- (c) (i) Using information from Table 2, list the corresponding amino acids coded by the mRNA in Fig. 2.2.

..... [1]

- (ii) Explain why each codon consists of three bases.

.....

.....

.....

.....

..... [2]

A ten amino acid long section of the polypeptide chain was partially hydrolysed to release oligopeptides shown in Fig. 2.3.

gly-ala-gly-val

gly-gly-lys

gly-val-ile-phe

phe-thr-gly-gly

Fig. 2.3

- (d) With reference to Fig. 2.3 work out the order of amino acids in the polypeptide.

..... [1]

- (e) (i) Explain why single base deletions in a coding DNA sequence are more likely to produce non-functional proteins than single base substitutions.

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..... [2]

- (ii) Cystic fibrosis (CF) results from base deletions in the CFTR gene. Explain why the consequence is different from usual single base deletions.

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..... [2]

[Total: 15]

- 3** The Middle East Respiratory Syndrome (MERS) is a fatal viral respiratory illness caused by a strain of coronavirus called MERS-CoV. Through its spike proteins, the virus infects cells in the respiratory tract in humans, leading to symptoms of fever, cough and shortness of breath, and eventually death.

Fig. 3.1 shows a model of the MERS-CoV.

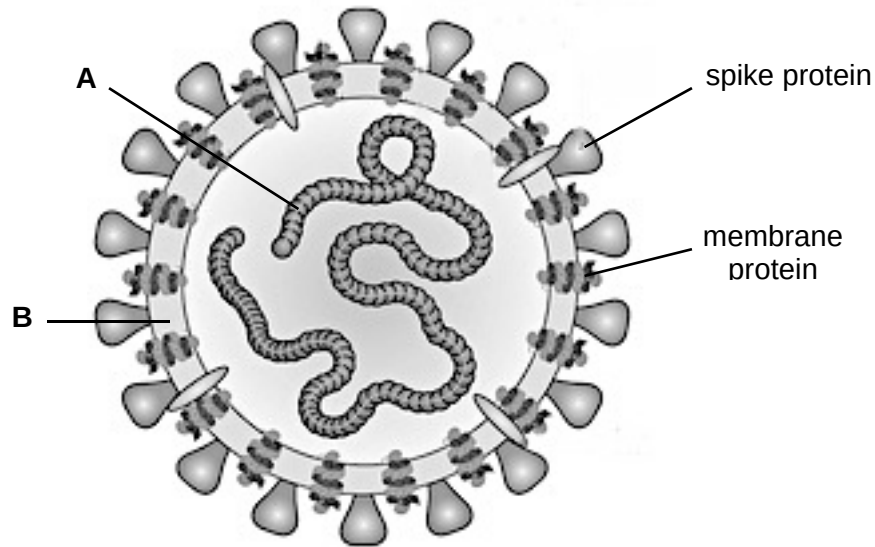


Fig. 3.1

- (a)** (i) Identify structure **A**.

[1]

- (ii) Explain the role of the positive sense RNA present in structure **A**.

[2]

- (iii) Describe how MERS-CoV acquires structure **B**.

[3]

- (b) A vaccine for MERS-CoV is to be developed in the near future. However, scientists warn that vaccines produced against the strain of MERS-CoV may not prevent future epidemics from occurring.

Explain why this is so.

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.....

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..... [3]

- (c) Unlike MERS-CoV, some human viruses are able to integrate their genome into the host cell genome. This sometimes leads to the development of cancer.

With a **named example**, explain how a virus may lead to the development of cancer.

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..... [4]

- (d) Scientists have sequenced the genome of MERS-CoV, and through bioinformatics analysis they have identified a low variable RNA segment in its genome. This allowed the synthesis of a short double-stranded RNA called the small interfering RNA (siRNA), which when introduced into the infected cell, is capable of preventing viral genome replication and protein synthesis.

Suggest how the siRNA is able to do so.

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..... [2]

[Total: 15]

- 4 A female carrier of X-linked SCID with blood type AB married a normal male with blood type O.

In blood transfusion, only individuals without genetic diseases are able to donate their blood.

- (a) Draw a genetic diagram to identify the offspring who are possible candidates for blood transfusion to their parents.

[5]

- (b) The ABO blood type shows discontinuous variation while body mass shows continuous variation in the human population.

Describe **two** differences between the two types of variation.

[2]

- (c) In an investigation, a mating between blood type AB and blood type O individuals was analysed. A chi-squared analysis was carried out and the chi-squared value calculated was 1.10. Using chi-squared values in Table 4, draw appropriate conclusions as to whether the results of the cross actually followed the expected ratio. Include in your conclusions a proposed null hypothesis.

Table 4

D.O.F	Probability, p				
	10%	5%	2%	1%	0.1%
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

----- [3]

[Total: 10]

- 5 Fig. 5 shows a neurone, neurone Z, forming synapses with adjacent neurones, neurone X and Y.

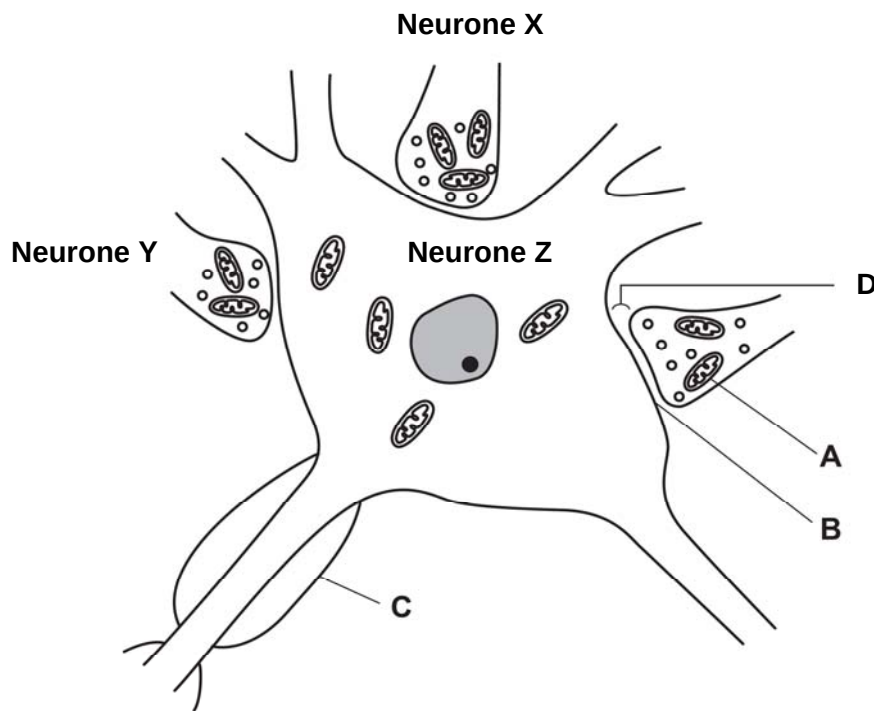


Fig. 5

- (a) (i) Name structures A to D.

A

B

C

D

[4]

- (ii) Outline the role of structure C in nerve impulse transmission.

.....

 [2]

- (b) Neurone **X** releases acetylcholine, a neurotransmitter which stimulates the production of nerve impulses in neurone **Z**. Neurone **Y** releases GABA, another neurotransmitter which results in the inhibition on the production of nerve impulses in Neurone **Z**.

- (i) Describe the sequence of events, from the arrival of a nerve impulse to neurone **X**, leading to the release of acetylcholine.

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..... [2]

- (ii) Explain how Neurone **Y** inhibits the production of nerve impulses in neurone **Z**.

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..... [2]

- (c) Epilepsy is a neurological disorder resulting in seizures when there is abnormal increase in neurone activity in the brain. Epilepsy has been linked to an abnormality in GABA receptors.

Suggest and explain how an abnormality in GABA receptors may result in epilepsy.

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..... [2]

[Total: 12]

- 6 The rosewood, *Tipuana tipu*, and the rambutan, *Nephelium lappaceum*, are some plants that share a common ancestor. Fig. 6 shows the variation in the pericarp (fruit wall) for a variety of methods of seed dispersal.

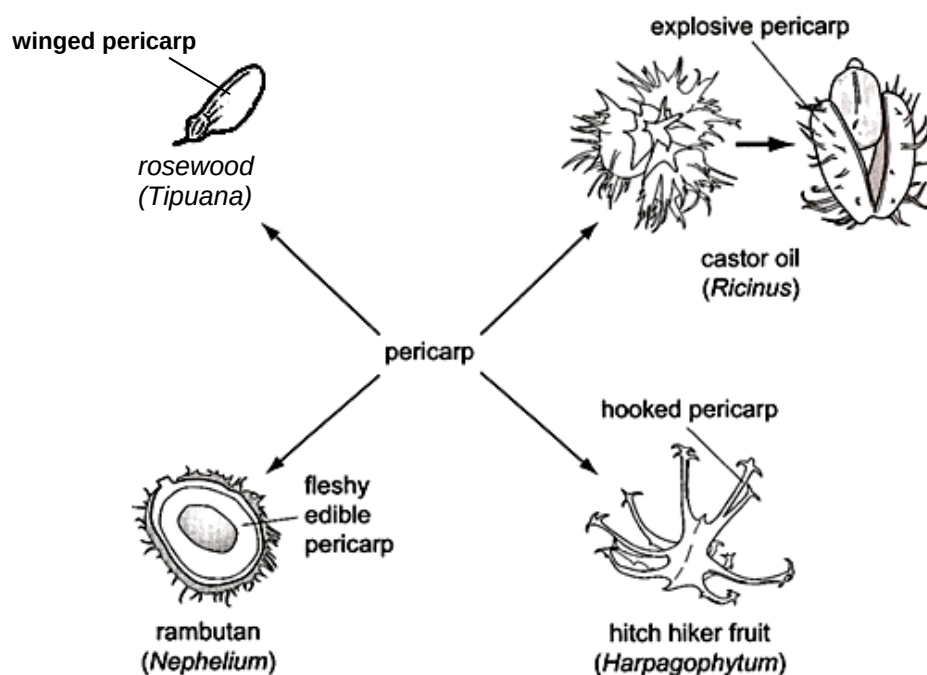


Fig. 6

- (a) Complete Table 6 to show the classification of the rosewood.

Table 6

kingdom	
	Streptophyta
	Magnoliopsida
order	Fabales
	Fabaceae
genus	
	<i>Tipuana tipu</i>

[3]

- 
- [5]

- [2]

- [4]

- (iii) In the case of extinct species, molecular homology may be less effective in determining evolutionary relationships.

Suggest why this is so.

----- [1]

[Total: 15]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section **(a)**, **(b)** etc., as indicated in the question.

- 7 (a) Describe the organisation of DNA in a chromosome during mitosis. [6]
- (b) Outline the events of mitosis. [8]
- (c) Describe the similarities and differences of homologous chromosomes. [6]

[Total: 20]

- 8** **(a)** Compare the endocrine and nervous system. [5]
- (b)** Describe how an action potential is transmitted along a myelinated neurone. [8]
- (c)** With reference to signal transduction, outline how insulin brings about a lowering of blood glucose concentration. [7]

[Total: 20]